

Enhancing Customer Experience and Brand Perception: Leveraging NLP and Digital Marketing for a Hotel in Porto

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Abstract. This paper presents a multi-stage NLP pipeline applied to 999 verified guest reviews collected from Booking.com for a boutique hotel in Porto, Portugal. The pipeline encompasses exploratory data analysis, supervised sentiment classification, unsupervised Latent Dirichlet Allocation (LDA) topic modelling, K-Means customer segmentation, and rule-based Aspect-Based Sentiment Analysis (ABSA). The overall average guest score is 8.24/10 across 38 months of data. The primary operational pain point identified by ABSA is WiFi & Check-in, where 39.5% of aspect mentions are negative. A binary LinearSVC classifier achieves F1-macro of 0.6471 (5-fold CV: 0.6939) and is proposed for real-time review monitoring. A fine-tuned XLM-RoBERTa transformer is evaluated as a multilingual proof-of-concept but underperforms classical baselines at this dataset size. Seven evidence-backed strategic recommendations with measurable KPIs are derived, addressing WiFi infrastructure, family guest experience, negative review response, staff-led marketing, and Iberian market expansion.

Keywords: Hotel reviews · Sentiment analysis · Topic modelling · Customer segmentation · ABSA · NLP · Digital marketing

1 Introduction

Online guest reviews represent one of the richest, most candid sources of intelligence available to hospitality operators. Unlike internal surveys, they are unprompted, public, and directly influence future booking decisions. For a hotel experiencing declining satisfaction scores and stagnating brand perception, systematically mining this corpus offers both a diagnostic and a strategic compass.

This paper presents the end-to-end findings of a data science pipeline applied to **999 verified guest reviews** collected from Booking.com between March 2023 and April 2026. The pipeline—built with LangGraph for orchestration and scikit-learn [1] for NLP modelling—covers exploratory analysis, supervised sentiment classification, unsupervised topic modelling, K-Means customer segmentation, and evidence-backed strategic recommendations.

The headline finding is an overall average score of **8.24/10**, masking significant variation by traveller segment and topic cluster. The following sections

unpack that variation and translate it directly into marketing and operational actions.

2 Data Collection and Preparation

2.1 Scraping and Ingestion

Reviews were collected programmatically from Booking.com using a custom Selenium-based scraper targeting a single boutique property in Porto, Portugal. The scraper handles dynamic pagination, modal dialogs, and deduplication via content-based fingerprinting. Each review record captures: reviewer name, country, room type, number of nights, date of stay, traveller type, positive and negative review text, and numerical score.

Data ethics note: Reviewer names and nationalities constitute personal data under GDPR. Names were collected solely for deduplication and are not used in any analysis. All data was collected from publicly accessible pages in compliance with Booking.com’s terms of service [5].

2.2 Cleaning and Feature Engineering

After automated cleaning, **999 reviews** were retained from an initial 1,000 (one dropped for empty review text). The pipeline applied: date parsing to period strings (YYYY-MM); number-of-nights extraction; concatenation of positive and negative text fields into a unified `full_review_text` column; language detection with `langdetect` and translation of non-English reviews via Google Translate; and derivation of a three-class sentiment label from the numerical score (positive: ≥ 8 ; neutral: 6–7; negative: < 6).

3 Exploratory Data Analysis

3.1 Score and Sentiment Distribution

Scores range from 1.0 to 10.0 on Booking.com’s scale, with a mean of **8.24** (median 8.0, std 1.68). The distribution is left-skewed: the majority of reviewers score the hotel 8 or above (Fig. 1). Sentiment labels derived from scores yield 779 positive (78.0%), 156 neutral (15.6%), and 64 negative (6.4%) reviews.

3.2 Temporal Trends

Review volume peaks in March 2026 (86 reviews). Stacked-area analysis of monthly sentiment share (Fig. 2) reveals a broadly stable positive share over 38 months, indicating no systematic operational deterioration. Short-term spikes in neutral and negative share correlate with periods of higher review volume, suggesting busy periods may strain service quality.

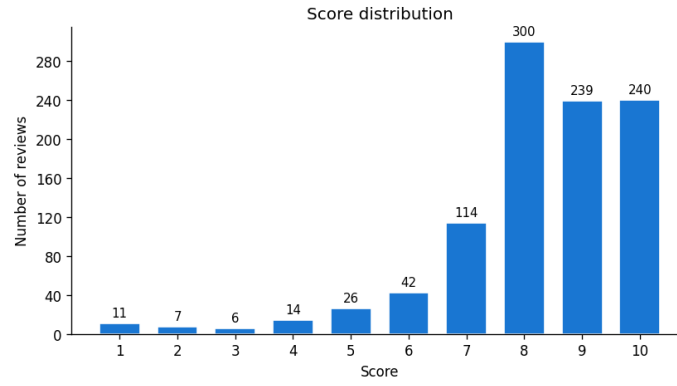


Fig. 1. Score frequency distribution across 999 reviews.

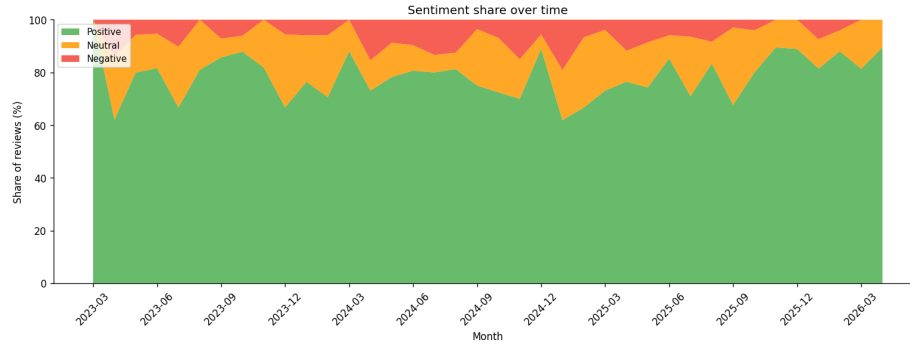


Fig. 2. Monthly sentiment share (stacked area) over 38 months.

3.3 Word Frequency

Word cloud analysis (Fig. 3) confirms that positive reviews centre on *location*, *staff*, *breakfast*, and *friendly*, while negative reviews surface *noise*, *small*, *air conditioning*, and *WiFi* as recurring pain points.

3.4 Guest Origin and Traveller Mix

The hotel draws an international audience dominated by the **United Kingdom** (238 reviews, 23.8%), followed by the United States (145) and Ireland (73). Iberian markets (Spain + Portugal: 89 reviews, 8.9%) are significantly underrepresented given the hotel’s Porto location—a strategic gap addressed in Section 8. Traveller type scores are shown in Table 1.

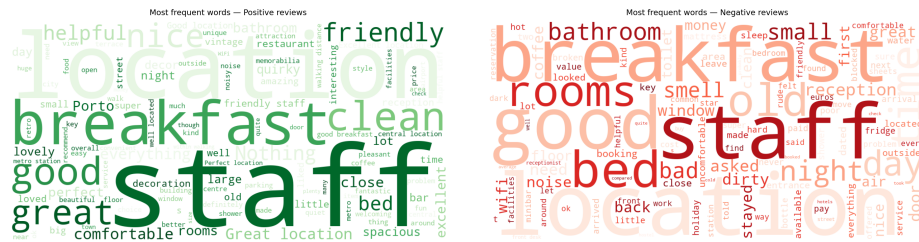


Fig. 3. Word clouds for positive reviews (left) and negative reviews (right).

Table 1. Average Booking.com score by traveller type.

Traveller type	Avg score	Reviews
Couple	8.35	519
Group	8.24	130
Solo traveller	8.13	226
Family	7.98	124

4 Customer Segmentation

K-Means clustering ($k = 4$) was applied to 999 reviews using five feature dimensions: guest score, review word count, number of nights, traveller type (one-hot encoded), and sentiment (ordinal). Features were standardised with StandardScaler. The value $k = 4$ was selected from an elbow analysis ($k = 2 \dots 8$) where inertia reduction flattened beyond four clusters.

Limitation: Clustering was performed at the review level, not the customer level. A single guest with multiple stays may appear across different clusters. Future work should aggregate features per unique reviewer.

The four segments are summarised in Table 2. **Family Explorers** carry the highest negative share (9.7%) and lowest average score, validating targeted improvements in Section 8. **Romantic Couples**—the largest, highest-scoring segment—represent the strongest base for loyalty campaigns.

5 Sentiment Analysis

5.1 Models and Approach

Three classifiers were evaluated against a VADER [2] lexicon baseline across two task formulations: 3-class (positive / neutral / negative) and binary (positive vs. non-positive). All supervised models use TF-IDF features (max 8,000 terms, 1–2 n-grams, sublinear TF scaling) with `class_weight='balanced'` to compensate for the 78/15.6/6.4% class imbalance. Evaluation uses macro-averaged F1. Cross-validation uses 5-fold stratified K-Fold.

Additionally, `xlm-roberta-base` [4] was fine-tuned for 2 epochs as a multilingual benchmark.

Table 2. K-Means customer segments ($k = 4$).

Segment	Size	Share	Avg score	Neg share
Romantic Couples	518	51.9%	8.35	5.2%
Solo Adventurers	226	22.6%	8.13	7.1%
Social Groups	131	13.1%	8.24	6.9%
Family Explorers	124	12.4%	7.98	9.7%

Table 3. Sentiment model results. CV F1 uses 5-fold stratified K-Fold.

Model	Task	Accuracy	F1-macro	CV F1-macro
VADER (baseline)	3-class	0.715	0.4511	—
TF-IDF + LR	3-class	0.740	0.5293	0.4909 ± 0.039
TF-IDF + LinearSVC	3-class	0.800	0.5344	0.4436 ± 0.043
TF-IDF + LinearSVC	Binary	0.750	0.6471	0.6939 ± 0.011
XLm-RoBERTa (fine-tuned)	3-class	0.780	0.2921	—

5.2 Results

Results are shown in Table 3 and Fig. 4. The 3-class macro F1 for LR (0.5293) reflects a structural *label-signal mismatch*: review text at the score-7/score-8 boundary is linguistically near-identical, making the neutral/negative boundary unreliable. Both supervised models nonetheless outperform the VADER baseline (0.4511).

5.3 XLm-RoBERTa Discussion

XLm-RoBERTa achieved F1-macro 0.2921—**below the VADER baseline**. This is expected: transformer models require substantially more labelled data than the ~ 800 training samples available here. With only 2 fine-tuning epochs, the model has insufficient exposure to hotel-review vocabulary. This experiment should be interpreted as a *proof-of-concept for the multilingual upgrade path*, not a performance improvement. At the current dataset size, TF-IDF + LinearSVC remains the recommended production approach.

5.4 Operational Application

The binary classifier (F1-macro 0.6471, CV 0.6939) is the most deployable model. Recommended use: **auto-flag incoming non-positive reviews (score < 8) within minutes of publication**, triggering a 24-hour management response SLA.

6 Topic Modelling

Latent Dirichlet Allocation (LDA) [3] was applied to all 999 review texts. The vocabulary was built with CountVectorizer (max 3,000 features, 1–2 n-grams,

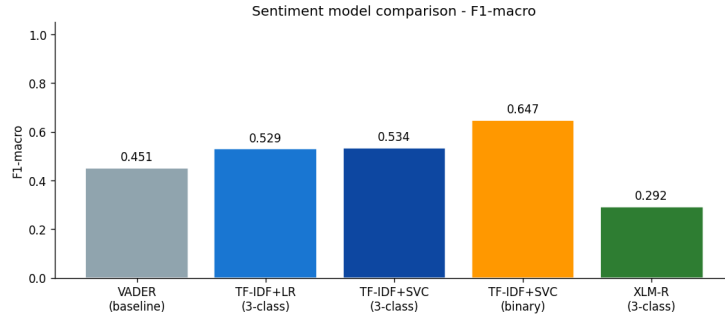


Fig. 4. F1-macro comparison across all sentiment models.

Table 4. LDA topics discovered ($k = 6$). Topics sorted by review count.

Topic label	Reviews	Avg score	Top keywords
Location & Accessibility	362	8.16	location, breakfast, staff, close, bathroom
Staff & Service	280	8.64	staff, location, friendly, helpful, clean
Food & Breakfast	219	8.32	location, breakfast, staff, excellent, perfect
Room Comfort & Cleanliness	112	7.36	parking, location, check, old, bed
Rooftop Bar & Ambiance	19	8.47	pillows, design, comfort, doors, extremely
Check-in & WiFi	7	7.43	hear, people, dirty, charged, buy

min_df=3, max_df=90%), yielding 1,289 unique terms after hotel-domain and multilingual stop-word filtering. Six topics were selected after comparing perplexity scores across $k \in \{4, 6, 8\}$; $k = 6$ produced the most interpretable separation while avoiding over-fragmentation at $k = 8$. Perplexity: **844.65**.

Discovered topics are shown in Table 4 and Fig. 5. **Location & Accessibility** is the dominant theme by volume (362 reviews). **Staff & Service** records the highest average score (8.64/10) and is the hotel’s primary reputation differentiator. **Room Comfort & Cleanliness** records the lowest score (7.36/10). The *Check-in & WiFi* topic (7 reviews) is statistically thin and should be interpreted with caution; the ABSA findings in Section 7 provide more robust evidence for this pain point.

7 Aspect-Based Sentiment Analysis

ABSA was applied using a rule-based approach: for each of eight hotel-domain aspects, keyword occurrences were located in review text and a ± 12 -word context window was scored with VADER. A total of **2,156 deduplicated aspect mentions** were extracted across all 999 reviews.

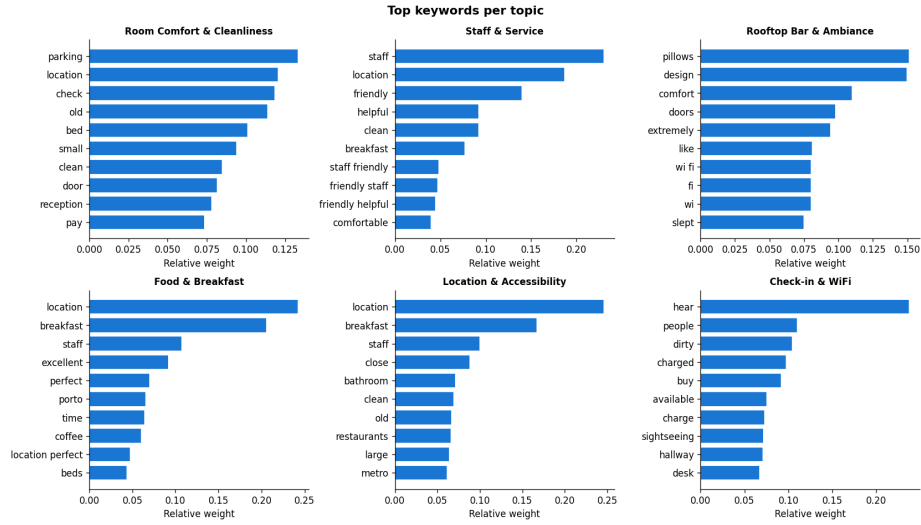


Fig. 5. Top keywords per LDA topic (relative weight).

Table 5. ABSA results sorted by negative mention share.

Aspect	Mentions Negative Neg %	
WiFi & Check-in	114	45 39.5%
Noise	99	32 32.3%
Room	444	96 21.6%
Breakfast	314	48 15.3%
Value	74	10 13.5%
Cleanliness	176	19 10.8%
Staff & Service	424	37 8.7%
Location	511	37 7.2%

Results are shown in Fig. 6 and Table 5. **WiFi & Check-in** is the highest pain point with **39.5% negative mentions** across 114 reviews. **Noise** follows at 32.3%. **Location** generates the most discussion (511 mentions) at only 7.2% negative—the hotel’s clearest marketing asset. The traveller $\times$ aspect heatmap (Fig. 7) reveals that Family guests show elevated negative rates for Room and Noise aspects relative to Couples and Solo travellers.

8 Strategic Recommendations

Seven evidence-backed recommendations are derived directly from the pipeline artifacts. Table 6 provides a decision-table summary with owners and timelines.

Key high-priority recommendations:

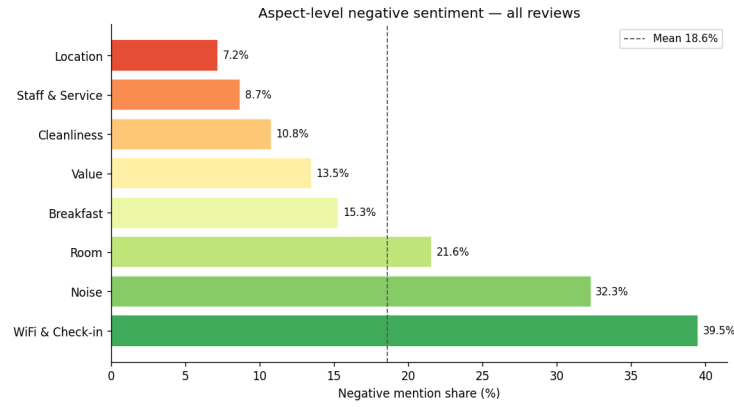


Fig. 6. Negative mention share (%) by aspect. Dashed line shows mean.

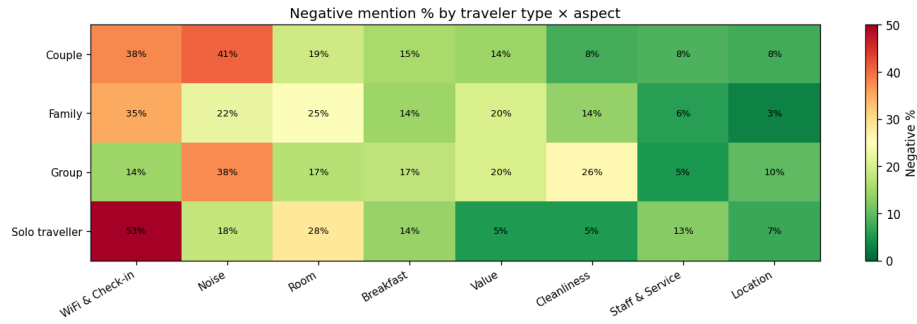


Fig. 7. Negative mention % by traveller type × aspect (heatmap).

R1 — WiFi & Check-in: Audit signal strength across all floors; install repeaters in weak zones; introduce digital key/PIN access to eliminate late-night lockout issues.

R2 — Family Experience: Offer interconnecting rooms and cot availability; add child-oriented breakfast items; distribute a printed city guide for families.

R4 — Negative Review Response: Deploy the binary sentiment classifier to auto-flag non-positive reviews within minutes of publication, triggering a 24-hour management response SLA.

R3 — Staff Marketing: Feature authentic guest quotes about staff in OTA listings; launch a ‘Staff Spotlight’ social media series; nominate consistently praised staff for hospitality awards.

Table 6. Strategic decision table. All KPIs measured within stated timeline.

ID	Finding	KPI	Timeline
R1	WiFi & Check-in: neg ABSA	39.5% Topic avg ≥ 8.50	0–6 mo
R2	Family score (low-est)	7.98 (low- Family avg ≥ 8.20)	0–9 mo
R4	6.4% negative, unsponded	Neg share $\leq 4\%$	0–3 mo
R3	Staff & Service avg	8.64 +10% direct bookings	3–12 mo
R5	Iberian share Porto	8.9% in Iberian share $\geq 15\%$	6–18 mo
R6	Room Comfort avg	7.36 Topic avg ≥ 8.40	3–9 mo
R7	Value aspect 13.5% neg	Value avg ≥ 8.60	3–6 mo

9 Conclusions

This analysis of 999 guest reviews delivers a clear, data-driven picture of hotel performance. An average score of 8.24/10 and 78.0% positive sentiment confirm that the hotel’s core proposition—friendly staff, unique vintage character, and central Porto location—resonates strongly with guests.

Three findings demand immediate operational attention: (1) WiFi and check-in friction anchors the lowest-scoring ABSA aspect (39.5% negative), fixable with targeted infrastructure investment; (2) Family Explorers are the lowest-scoring segment (7.98/10), representing an underserved market; and (3) 6.4% of reviews are negative—a manageable number that the binary classifier (F1-macro 0.6471) can surface in real time.

The hotel’s greatest strength—staff quality—should be leveraged aggressively in OTA copy, social media, and PR to reinforce brand differentiation and justify premium pricing. Medium-term growth lies in Iberian market expansion (currently 8.9% of reviews), converting Romantic Couples into repeat bookers via a loyalty programme, and standardising room comfort across all room types.

Limitations

- *Label-signal mismatch*: Score-derived sentiment labels conflate linguistically similar positive/neutral reviews, capping 3-class F1-macro at ~ 0.53 .
- *Multilingual corpus*: Non-English reviews introduce noise into LDA keyword lists despite stop-word filtering.
- *Single property*: Findings are specific to one hotel; cross-property benchmarking would strengthen recommendation confidence.
- *Review-level clustering*: K-Means segments reviews, not customers; future work should aggregate per unique reviewer.

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